condition: separating runs alone would still let two runs of the same plant fall on opposite sides of the boundary and turn a claim about unseen plants into a claim about unseen episodes.

The strict open-loop export needs a different treatment because it contains only four runs and holding one out would remove an entire plant draw from training. Instead each run is divided into twenty temporal blocks, every fifth block is assigned to validation, and a five-sample guard interval is discarded at each block boundary so that no 0.05 s transition window spans the split. A simple last-twenty-percent split was tried first and rejected: it changed the spread of the targets between training and validation by a factor of 1.3 to 2.3 because the final fifth of the run is the simultaneous excitation phase and is dynamically unlike everything before it. That split would have measured extrapolation to a different regime and called it generalisation.

Normalization statistics are computed on the training split only and stored inside each checkpoint, so that a model always normalizes with the statistics it was fitted with. Mixing statistics between checkpoints silently invalidates any comparison, and the loader enforces the stored contract rather than recomputing it.

Finally, samples are carried with an explicit validity flag rather than being silently dropped In the strict export 93.7 % of samples are marked valid. The remainder are the excluded startup and shutdown seconds and the zero-command gaps. Keeping them in the archive with a flag, instead of deleting them, means the time base stays contiguous and the within-run difference checks remain meaningful.

4.10 Dataset Diagnostics

No dataset in this thesis is used before it passes an explicit audit, for the reason given at the end of Section 4.4. The audit of the strict open-loop export is reproduced in Table B.4

The two rank figures are the ones that speak directly to requirement R2. The instantaneous command matrix has rank four, so no channel is a linear combination of the others, and the lagged regressor matrix built from the command history has rank eighty, so the excitation is persistently exciting to the order the models need. The condition number of that lagged matrix is about 4.6×10^3, which is finite and workable rather than comfortable. It is a consequence of using long plateaus, and it is the price paid for the stroke coverage those plateaus buy.

The exactness of the three increment checks deserves a word, since a maximum error of exactly zero invites suspicion. It is expected here: the targets are computed from the logged states by the same within-run difference that the check recomputes, so the check verifies that the extraction stage did what it claims, not that the simulation is accurate. It catches misalignment, off-by-one indexing, and rows that have been reordered or concatenated across run boundaries, which are the failure modes that occur. Physical fidelity is a separate question, addressed by the validation study in Section 3.4.4.

Two limitations of the Simscape material remain. Stroke coverage reaches only 12 % on the head-pivot cylinder, which is also the cylinder the strict forward model predicts least well The rod-force channel fails a piston-area consistency check on this export, so it carries zero supervised weight in the reduced-order model and no result rests on it.